CS341 · COMPUTER GRAPHICS

Mock Midterm Solutions

20 marks · 1 hour · same format as the past midterms

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01 Solution Q1

1. The **frame buffer** (refresh buffer, bit map) is the memory area that stores the picture of a raster system; each location corresponds to one pixel.
2. **Persistence** is the time for the emitted light to decay to one-tenth of its original intensity. **Low** persistence is better for animation.

3) $v = Q - P = (1 - 4,\ 5 - 1) = (-3,\ 4)$ $\quad |v| = \sqrt{(9 + 16)} = 5$

4) $$AR = \frac{\frac{720}{6}}{\frac{1280}{10}} = \frac{120}{128} = 0.9375$$

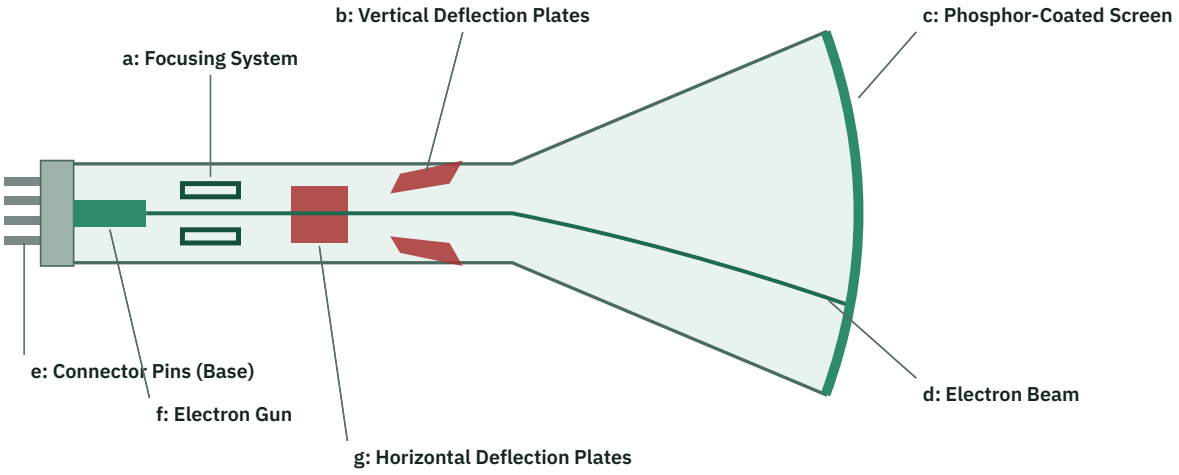
02 Solution Q2

#	Answer	Reason
1	**True**	
2	**False**	Plasma is emissive; LCD is the nonemissive one.
3	**True**	
4	**True**	
5	**True**	
6	**True**	And x increases by $\frac{1}{m}$.
7	**False**	To one-tenth, not half.
8	**False**	The result is a scalar.

03 Solution Q3

1. b 2. c 3. b 4. c 5. b 6. b 7. b 8. b

04 Solution Q4



a: Focusing System b: Vertical Deflection Plates c: Phosphor-Coated Screen d: Electron Beam

e: Connector Pins (Base) f: Electron Gun g: Horizontal Deflection Plates

05 Solution Q5

$$dx = 4, \quad dy = 8, \quad m = \frac{8}{4} = 2 > 1$$

$$steps = max(|dx|, |dy|) = 8$$

$$xInc = \frac{4}{8} = 0.5 \qquad yInc = \frac{8}{8} = 1$$

$$round(a) = int(a + 0.5)$$

k	x	y	round(x)	round(y)
0	1	2	1	2
1	1.5	3	2	3
2	2	4	2	4
3	2.5	5	3	5
4	3	6	3	6
5	3.5	7	4	7
6	4	8	4	8
7	4.5	9	5	9
8	5	10	5	10

